Systems of two linear equations in two variables — Part 1 answer key

#	ID	Answer	Difficulty
1.1	026980-DC	?	Easy
1.2	10239-DC	B	Easy
1.3	4b320a23-0ce8-4391-9201-ef14dec4437c	C	Easy
1.4	027878-DC	C	Easy
1.5	a90c85c1-804c-489e-9c12-436ca82f1f2d	4	Easy
1.6	7eab7aa7-dffb-49ed-89bb-e363db1aaad3	C	Easy
1.7	ac10f741-d7d2-4630-ac36-63d9fd1f7a94	A	Easy
1.8	ebe78c3d-d888-4a7c-8973-f50c7ec643f3	C	Easy
1.9	09904-DC	A	Easy
1.10	027122-DC	B	Easy
1.11	2afd7270-62a5-4b7b-9b32-31529be1acdd	C	Easy
1.12	07542-DC	B	Easy
1.13	08822-DC	B	Easy
1.14	f0975e35-afbb-4d74-9091-7a06a2bf4d03	A	Easy
1.15	632affe2-9dad-466c-b099-d98aa7c13b20	C	Easy
1.16	96df58ca-4a03-48f8-b5e1-0437cd8a8511	B	Easy
1.17	52842d0a-65a9-4a31-89c5-5df2054afc60	B	Easy
1.18	67838907-7e84-4f9a-bb16-0c54fe5e5232	30	Easy
1.19	f0dd503e-886e-49ec-9002-e65200346c57	6	Easy
1.20	0fe484a0-8389-4fe5-93dd-5d56c07a7c9c	D	Easy

1.21	950dd818-fa5e-4231-b816-a13da78257e9	C	Easy
1.22	08086-DC	A	Easy
1.23	648f6724-1596-45f8-99c7-238a1a0648f9	B	Easy
1.24	fef8ab08-2a09-4477-9b15-1ed3a9fc0ac6	A	Easy
1.25	026034-DC	B	Easy
1.26	d9637287-6236-4171-8284-30cfa0ba9e23	A	Easy
1.27	9848a7b9-5a10-4833-b6e0-1b98ce1e44c1	D	Easy
1.28	020079-DC	A	Easy
1.29	f55e2a62-1692-407c-9001-96da7e84f1bb	D	Easy
1.30	ff17c197-f425-42d5-b68e-437f6191b533	A	Easy
1.31	026145-DC	A	Easy
1.32	2b87b482-ed79-4f52-87e1-3cd4aec4b340	B	Easy
1.33	f0db62a7-74c8-4e51-91dd-8c8b80649a49	A	Easy
1.34	08084-DC	B	Easy
1.35	58253567-9086-41df-86cd-17a48bbb725b	B	Easy
1.36	5f351eb6-67ee-4ee2-a604-17867c3570b7	A	Easy
1.37	680ee560-0171-401e-8a31-87727153a417	B	Easy
1.38	55751ff1-1079-45a7-b117-348b4d276815	B	Easy
1.39	c470d4b9-a5d3-44e7-9e33-d10b5151dcde	D	Medium
1.40	07807ef4-fe53-474a-92e4-59c6ea6c579e	B	Medium
1.41	ac648dfb-04b8-480f-bc41-fe111d3c500c	1	Medium
1.42	08687-DC	D	Medium
1.43	5c64450d-1ff5-486d-bc37-e47fa80d2f7d	A	Medium
1.44	a94ddb79-a5ea-48b4-ac01-838bf92d16fb	25	Medium
1.45	022962-DC	C	Medium
1.46	6224ef79-ede3-42e9-82fc-447127d5e27e	A	Medium

1.47	dce39b31-9110-4b75-93ff-5d7da57a6adc	C	Medium
1.48	4d15e7be-e330-43ce-90bb-537310380c04	C	Medium
1.49	1fe5014a-3025-473b-a798-53f9ceab24bd	C	Medium
1.50	9074f61c-00b0-4492-ab66-501db8f46f38	D	Medium
1.51	14931-DC	C	Medium
1.52	74ba4b49-0d96-4325-b90c-ffaf55d59428	B	Medium
1.53	4754b73c-5377-41de-93c1-d49dd784c2c0	40	Medium
1.54	20220edf-4df0-47cb-bce6-798d01c48907	B	Medium
1.55	b1b506b9-ecab-4a7d-9043-cf9ca8ba8099	3	Medium
1.56	4e08d101-c21c-44f2-a0d1-b1575f5b4632	C	Medium
1.57	a18ef7d1-8bbd-459a-93cc-e4df62df2c1a	B	Medium
1.58	f9cb4504-e307-4fd6-973f-1bbab97daf43	C	Medium
1.59	b0266b5e-6f26-496c-81ea-3108b31c0bc6	D	Medium
1.60	1ce8eb76-74a1-43fe-a9d0-df147e4ca59e	A	Medium
1.61	05f30aab-94fb-4835-aaaf-c316bafa965e	A	Medium
1.62	7bbaeebb-c7b9-4d87-ac22-9fb4f0d77968	24	Medium
1.63	73c4251e-9b52-43e0-b64c-eafae656c149	D	Medium

Rationales

▼ 1.1 — ?

The correct answer is 23. Since it's given that $x = 2$, the value of y can be found by substituting 2 for x in the first equation and solving for y. Substituting 2 for x yields $3(2) + y = 29$, or $6 + y = 29$. Subtracting 6 from both sides of this equation yields $y = 23$.

▼ 1.2 — B

Choice B is correct. Let n be the number of novels and m be the number of magazines that Sadie purchased. If Sadie purchased a total of 11 novels and magazines, then $n + m = 11$. It

is given that the combined price of 11 novels and magazines is $20. Since each novel sells for $4 and each magazine sells for $1, it follows that $4n+m=20$. So the system of equations below must hold.

$$\begin{aligned} 4n+m &= 20 \\ n+m &= 11 \end{aligned}$$

Subtracting corresponding sides of the second equation from the first equation yields $3n=9$, so $n=3$. Therefore, Sadie purchased 3 novels.

Choice A is incorrect. If 2 novels were purchased, then a total of $8 was spent on novels. That leaves $12 to be spent on magazines, which means that 12 magazines would have been purchased. However, Sadie purchased a total of 11 novels and magazines. Choices C and D are incorrect. If 4 novels were purchased, then a total of $16 was spent on novels. That leaves $4 to be spent on magazines, which means that 4 magazines would have been purchased. By the same logic, if Sadie purchased 5 novels, she would have no money at all ($0) to buy magazines. However, Sadie purchased a total of 11 novels and magazines.

▼ 1.3 — C

Choice C is correct. It's given that admission to the museum will cost $\$8.50$ for each student and $\$12$ for each chaperone. It's also given that it will cost a total of $\$730$ for admission to the museum for all 83 people on the field trip. Let x represent the number of students and y represent the number of chaperones going on the field trip. Since there are a total of 83 people going on the field trip, it follows that $x+y=83$. Since admission to the museum for each student costs $\$8.50$ and each chaperone costs $\$12$, the total cost can be represented by the equation $8.50x+12y=730$. The equations $x+y=83$ and $8.50x+12y=730$ form a system of equations. Subtracting x from both sides of the equation $x+y=83$ yields $y=83-x$. Substituting $83-x$ for y in the equation $8.50x+12y=730$ yields $8.50x+12(83-x)=730$, or $-3.50x+996=730$. Subtracting 996 from both sides of this equation yields $-3.50x=-266$. Dividing both sides of this equation by -3.50 yields $x=76$. Therefore, 76 students are going on the field trip.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.4 — C

Choice C is correct. A system of two linear equations has infinitely many solutions if one equation is equivalent to the other. This means that when the two equations are written in the same form, each coefficient or constant in one equation is equal to the corresponding coefficient or constant in the other equation multiplied by the same number. The equations in the given system of equations are written in the same form, with x and y on the left-hand side of the equation and a constant on the right-hand side of the equation. The coefficients of x and y in the second equation are equal to the coefficients of x and y, respectively, in the first equation multiplied by 4: $8 = 2(4)$ and $28 = 7(4)$. Therefore, the constant in the second equation must be equal to 4 times the constant in the first equation: $a = 9(4)$, or $a = 36$.

Choices A, B, and D are incorrect. When $a = 4$, $a = 9$, or $a = 54$, the given system of equations has no solution.

▼ 1.5 — 4

The correct answer is 4. A solution to a system of equations must satisfy each equation in the system. It follows that if x, y is a solution to the system, the point x, y lies on the graph in the xy-plane of each equation in the system. According to the graph, the point x, y that lies on the graph of each equation in the system is $4, 1$. Therefore, the solution to the system is $4, 1$. It follows that the value of x is 4.

▼ 1.6 — C

Choice C is correct. The given system of linear equations can be solved by the substitution method. Substituting $-3x$ for y from the first equation in the given system into the second equation yields $4x + -3x = 15$, or $x = 15$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the absolute value of y, not the value of x.

▼ 1.7 — A

Choice A is correct. The second equation in the given system is $x = 8$. Substituting 8 for x in the first equation in the given system yields $4(8) - 3y = 5$, or $32 - 3y = 5$. Subtracting 32 from both sides of this equation yields $-3y = -27$. Dividing both sides of this equation by -3 yields $y = 9$. Therefore, the solution (x, y) to the given system of equations is $(8, 9)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.8 — C

Choice C is correct. The solution to a system of linear equations is represented by the point that lies on the graph of each equation in the system, or the point where the lines intersect on a graph. On the graph shown, the two lines intersect at the point $-4, -3$. Therefore, the solution to the system is $-4, -3$.

Choice A is incorrect. This is the y-intercept of the graph of one of the lines shown, not the intersection point of the two lines.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.9 — A

Choice A is correct. The airline sold two types of tickets for this flight: x tickets at $90 each and the remaining tickets, y, at $250 each. Because the airline sold a total of 120 tickets for this flight, it must be true that x + y = 120. The amount, in dollars, collected from the sale of x tickets at $90 each is represented by 90x. The amount, in dollars, collected from the sale of the remaining y tickets at $250 each is represented by 250y. It is given that a total of $27,600 was collected from the sale of all tickets. Therefore, it must also be true that 90x + 250y = 27,600.

Choice B is incorrect. The total number of tickets sold is represented correctly as x + y = 120. The total amount, in dollars, collected from the sale of the x tickets at $90 each and the remaining tickets, y, at $250 has been correctly represented as 90x + 250y. However, according to the information given, this total should be equal to 27,600, not 120(27,600) dollars. Choice C is incorrect. The total number of tickets sold has been correctly represented as x + y. However, according to the information given, this total should be equal

to 120, not 27,600, as shown in choice C. The total amount, in dollars, collected from the sale of the x tickets at $90 each and the remaining tickets, y, at $250 has been correctly represented as 90x + 250y. However, according to the information given, this total should be equal to 27,600, not 120(27,600) dollars. Choice D is incorrect. The two equations given in choice D have no meaning in this context.

▼ 1.10 — B

Choice B is correct. Since it's given that $x = 1$, substituting 1 for x in the first equation yields $y = 2(1) + 3$. Simplifying the right-hand side of this equation yields $y = 2 + 3$, or $y = 5$. Therefore, the ordered pair $(1,5)$ is a solution to the given system of equations.

Choice A is incorrect and may result from a calculation error when substituting 1 for x in the first equation. Choices C and D are incorrect. Because it's given that $x = 1$, x cannot equal 2 as stated in these ordered pairs.

▼ 1.11 — C

Choice C is correct. It's given that $4x = 20$ and $-3x + y = -7$ is a system of equations with a solution x, y. Adding the second equation in the given system to the first equation yields $4x + (-3x + y) = 20 + -7$, which is equivalent to $x + y = 13$. Thus, the value of $x + y$ is 13.

Choice A is incorrect. This represents the value of $-2x + y - 1$.

Choice B is incorrect. This represents the value of $-x + y$.

Choice D is incorrect. This represents the value of $2x + y + 1$.

▼ 1.12 — B

Choice B is correct. The movie theater sells f full-price tickets and r reduced-price tickets, so the total number of tickets sold is f + r. Since the movie theater sold a total of 214 full-price and reduced-price tickets for one movie showing, it follows that f + r = 214. The movie theater charges $11 for each full-price ticket; thus, the sales for full-price tickets, in dollars, is given by 11f. The movie theater charges $8.25 for each reduced-price ticket; thus, the sales for reduced-price tickets, in dollars, is given by 8.25r. Therefore, the total sales, in dollars, for the movie showing is given by 11f + 8.25r. Since the total sales for all full-price and reduced-price tickets is $2,145, it follows that 11f + 8.25r = 2,145.

Choice A is incorrect. This system of equations suggests that the movie theater sold a total of 2,145 full-price and reduced-price tickets for a total of $214. Choice C is incorrect. This system suggests that the movie theater charges $8.25 for each full-price ticket and $11 for each reduced-price ticket. Choice D is incorrect. This system suggests that the movie theater charges $8.25 for each full-price ticket and $11 for each reduced-price ticket and sold a total of 2,145 tickets for a total of $214.

▼ 1.13 — B

Choice B is correct. The number of coins collected can be multiplied by c to give the score from the points earned from coins. Similarly, the number of stars collected can be multiplied by s to give the score from the points earned from the stars. Therefore, the total score each time Angela played is $20c + 10s = 700$, and the total score the second time she played is $25c + 12s = 850$.

Choices A, C, and D are incorrect and may result from misidentifying the terms of the equation. Choice A switches coins and stars, choice C switches stars and points, and choice D misidentifies coins, stars, and points.

▼ 1.14 — A

Choice A is correct. The second equation in the given system defines the value of x as $5y$. Substituting $5y$ for x into the first equation yields $5y + y = 18$ or $6y = 18$. Dividing each side of this equation by 6 yields $y = 3$. Substituting 3 for y in the second equation yields $53 = x$ or $x = 15$. Therefore, the solution x, y to the given system of equations is $15, 3$.

Choice B is incorrect. Substituting 16 for x and 2 for y in the second equation yields $52 = 16$, which is not true. Therefore, $16, 2$ is not a solution to the given system of equations.

Choice C is incorrect. Substituting 17 for x and 1 for y in the second equation yields $51 = 17$, which is not true. Therefore, $17, 1$ is not a solution to the given system of equations.

Choice D is incorrect. Substituting 18 for x and 0 for y in the second equation yields $50 = 18$, which is not true. Therefore, $18, 0$ is not a solution to the given system of equations.

▼ 1.15 — C

Choice C is correct. Adding the second equation of the given system to the first equation yields $5x + -4x + y = 15 + -2$, which is equivalent to $x + y = 13$. So the value of $x + y$ is 13.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $-(x + y)$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.16 — B

Choice B is correct. The second equation in the given system is $r = 3$. Substituting 3 for r in the first equation in the given system yields $s + 73 = 27$, or $s + 21 = 27$. Subtracting 21 from both sides of this equation yields $s = 6$. Therefore, the solution r, s to the given system of equations is $3, 6$.

Choice A is incorrect. This is the solution s, r, not r, s, to the given system of equations.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.17 — B

Choice B is correct. A solution to a system of equations must be the solution to each equation in the system. It follows that if x, y is a solution to the system, the point x, y lies on the graph in the xy-plane of each equation in the system. The point that lies on each graph of the system of linear equations shown is their intersection point $2, 4$. Therefore, the solution to the system is $2, 4$.

Choice A is incorrect. The point $0, 5$ lies on one, but not both, of the graphs of the linear equations shown.

Choice C is incorrect. The point $5, 10$ lies on one, but not both, of the graphs of the linear equations shown.

Choice D is incorrect. The point $10, 0$ lies on one, but not both, of the graphs of the linear equations shown.

▼ 1.18 — 30

The correct answer is 30. It's given that $x + y = 125$. Substituting 125 for $x + y$ in the second equation of the given system yields $125 + y = 155$. Subtracting 125 from both sides of this equation yields $y = 30$. Therefore, the value of y is 30.

▼ 1.19 — 6

The correct answer is 6. The first equation in the given system is $x = 8$. Substituting 8 for x in the second equation in the given system yields $8 + 3y = 26$. Subtracting 8 from both sides of this equation yields $3y = 18$. Dividing both sides of this equation by 3 yields $y = 6$. Therefore, the value of y is 6.

▼ 1.20 — D

Choice D is correct. It's given by the first equation in the given system of equations that $x = 10$. Substituting 10 for x in the second equation in the given system yields $y = 10 + 21$, or $y = 31$. Therefore, the value of y is 31.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x, not the value of y.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.21 — C

Choice C is correct. The solution to this system of linear equations is represented by the point that lies on both lines shown, or the point of intersection of the two lines. According to the graph, the point of intersection occurs when $x = 2$ and $y = 3$, or at the point $2, 3$. Therefore, the solution x, y to the system is $2, 3$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.22 — A

Choice A is correct. Hiro purchased 4 shirts and each shirt cost x dollars, so he spent a total of 4x dollars on shirts. Likewise, Hiro purchased 2 pairs of pants, and each pair of pants cost y dollars, so he spent a total of 2y dollars on pants. Therefore, the total amount that Hiro spent was 4x + 2y. Since Hiro spent $86 in total, this can be modeled by the equation

4x + 2y = 86. Using the same reasoning, Sofia bought 3 shirts at x dollars each and 5 pairs of pants at y dollars each, so she spent a total of 3x + 5y dollars on shirts and pants. Since Sofia spent $166 in total, this can be modeled by the equation 3x + 5y = 166.

Choice B is incorrect and may be the result of switching the number of shirts Sofia purchased with the number of pairs of pants Hiro purchased. Choice C is incorrect and may be the result of switching the total price each person paid. Choice D is incorrect and may be the result of switching the total price each person paid as well as switching the number of shirts Sofia purchased with the number of pairs of pants Hiro purchased.

▼ 1.23 — B

Choice B is correct. It's given by the second equation in the system that $y = 19$. Substituting 19 for y in the first equation yields $19 = 4x - 9$. Adding 9 to both sides of this equation yields $28 = 4x$. Dividing both sides of this equation by 4 yields $7 = x$. Therefore, since $x = 7$ and $y = 19$, the solution x, y to the given system of equations is $7, 19$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.24 — A

Choice A is correct. The solution to this system of linear equations is represented by the point that lies on both lines shown, or the point of intersection of the two lines. According to the graph, the point of intersection occurs when $x = 4$ and $y = -5$, or at the point $4, -5$. Therefore, the solution x, y to the system is $4, -5$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.25 — B

Choice B is correct. Substituting 20 for $x+y$ in the second equation in the system yields $2(20)+3y=85$, or $40+3y=85$. Subtracting 40 from both sides of this equation yields $3y=45$. Dividing both sides of this equation by 3 yields $y=15$.

Choice A is incorrect. If $y=10$, then $x=10$ since $x+y=20$. However, substituting 10 for both x and y in the second equation yields $70=85$, which is a false statement. Choice C is incorrect. If $y=60$, then $x=-40$ since $x+y=20$. However, substituting these values for x and y in the second equation yields $220=85$, which is a false statement. Choice D is incorrect. If $y=65$, then $x=-45$ since $x+y=20$. However, substituting these values for x and y in the second equation yields $235=85$, which is a false statement.

▼ 1.26 — A

Choice A is correct. The second equation in the given system is $y = 28$. Substituting 28 for y in the first equation in the given system yields $28 = 12x - 20$. Adding 20 to both sides of this equation yields $48 = 12x$. Dividing both sides of this equation by 12 yields $4 = x$. Therefore, the solution x, y to the given system of equations is $4, 28$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the solution y, x, not x, y, to the given system of equations.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.27 — D

Choice D is correct. Dividing both sides of the equation $10x = 110$ by 10 yields $x = 11$. Substituting 11 for x in the equation $6x - 63 = y$ yields $6(11) - 63 = y$, which is equivalent to $66 - 63 = y$, or $3 = y$. Therefore, the value of y is 3.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x, not y.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.28 — A

Choice A is correct. It's given that the petting zoo sells two types of tickets, standard and premium, and that s represents the number of standard tickets sold and p represents the number of premium tickets sold. It's also given that the petting zoo sold 250 tickets on one Saturday; thus, $s+p=250$. It's also given that each standard ticket costs $5 and each premium ticket costs $12. Thus, the amount collected in ticket sales can be represented by $5s$ for standard tickets and $12p$ for premium tickets. On that Saturday the petting zoo collected a total of $2,300 from ticket sales; thus, $5s+12p=2,300$. These two equations are correctly represented in choice A.

Choice B is incorrect. The second equation in the system represents the cost per standard ticket as $12, not $5, and the cost per premium ticket as $5, not $12. Choices C and D are incorrect. The equations represent the total collected from standard and premium ticket sales as $250, not $2,300, and the total number of standard and premium tickets sold as $2,300, not $250. Additionally, the first equation in choice D represents the cost per standard ticket as $12, not $5, and the cost per premium ticket as $5, not $12.

▼ 1.29 — D

Choice D is correct. It's given that the system has infinitely many solutions. The graphs of two lines in the *xy*-plane represented by equations in slope-intercept form, $y = mx + b$, where m and b are constants, have infinitely many solutions if their slopes, m, are the same and if their *y*-coordinates of the *y*-intercepts, b, are also the same. The first equation in the given system is $y = \frac{2}{7}x + 3$. For this equation, the slope is $\frac{2}{7}$ and the *y*-coordinate of the *y*-intercept is 3. If the second equation is in the form $y = mx + b$, then for the two equations to be equivalent, the values of m and b in the second equation must equal the corresponding values in the first equation. Therefore, the second equation must have a slope, m, of $\frac{2}{7}$, and a *y*-coordinate of the *y*-intercept, b, of 3. Thus, the value of b is 3.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

▼ 1.30 — A

Choice A is correct. The first equation in the given system of equations is $x = 4$. Substituting 4 for x in the second equation in the given system of equations yields $y = 5 - 4$, or $y = 1$.

Choice B is incorrect. This is the value of x in the solution to the given system of equations, not the value of y.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.31 — A

Choice A is correct. The solution to this system of linear equations is the point that lies on both lines graphed, or the point of intersection of the two lines. According to the graphs, the point of intersection occurs when $x = 3$ and $y = 9$, or at the point $(3,9)$.

Choices B and D are incorrect. Each of these points lies on one line, but not on both lines in the xy-plane. Choice C is incorrect. This point doesn't lie on either of the lines graphed in the xy-plane.

▼ 1.32 — B

Choice B is correct. If a point x, y lies on both lines in the graph of a system of two linear equations, the ordered pair x, y is a solution to the system. The graph shown is the graph of a system of two linear equations, where the two lines in the graph intersect at the point $3, 4$. Therefore, the point $3, 4$ lies on both lines, so the ordered pair $3, 4$ is the solution to the system.

Choice A is incorrect. The point $2, 3$ lies on one, not both, of the lines in the graph shown.

Choice C is incorrect. The point $4, 5$ lies on one, not both, of the lines in the graph shown.

Choice D is incorrect. The point $5, 6$ lies on one, not both, of the lines in the graph shown.

▼ 1.33 — A

Choice A is correct. It's given that Connor has c dollars, Maria has m dollars, and Connor has 4 times as many dollars as Maria. This can be represented by the equation $c = 4m$. It's also given that together, Connor and Maria have a total of $\$ 25.00$, which can be represented by the equation $c + m = 25$. Therefore, the system consisting of the equations $c = 4m$ and $c + m = 25$ represents this situation.

Choice B is incorrect. The equation $m = 4c$ represents a situation where Maria has 4 times as many dollars as Connor, rather than the situation where Connor has 4 times as many dollars as Maria.

Choice C is incorrect. The equation $c = 25m$ represents a situation where Connor has 25 times, rather than 4 times, as many dollars as Maria. The equation $c + m = 4$ represents a situation where Connor and Maria together have a total of $\$4.00$, rather than $\$25.00$.

Choice D is incorrect. The equation $m = 25c$ represents a situation where Maria has 25 times as many dollars as Connor, rather than the situation where Connor has 4 times as many dollars as Maria. The equation $c + m = 4$ represents a situation where Connor and Maria together have a total of $\$4.00$, rather than $\$25.00$.

▼ 1.34 — B

Choice B is correct. Outfits for boys cost $26 each and the teacher ordered b outfits for boys, so the teacher spent 26b dollars on outfits for boys. Similarly, outfits for girls cost $35 each and the teacher ordered g outfits for girls, so the teacher spent 35g dollars on outfits for girls. Since the teacher spent a total of $881 on outfits for boys and girls, the equation 26b + 35g = 881 must be true. And since the teacher ordered a total of 28 outfits, the equation b + g = 28 must also be true.

Choice A is incorrect and may result from switching the constraint on the total number of outfits with the constraint on the cost of the outfits. Choice C is incorrect and may result from switching the constraint on the total number of outfits with the constraint on the cost of the outfits, as well as switching the cost of the outfits for boys with the cost of the outfits for girls. Choice D is incorrect and may result from switching the cost of the outfits for boys with the cost of the outfits for girls.

▼ 1.35 — B

Choice B is correct. The solution to this system of linear equations is represented by the point that lies on both lines shown, or the point of intersection of the two lines. According to the graph, the point of intersection occurs when $x = 2$ and $y = 2$, or at the point $2, 2$. Therefore, the solution x, y to the system is $2, 2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.36 — A

Choice A is correct. According to the first equation in the given system, $y = 4$. Substituting 4 for y in the second equation in the given system yields $x = 4 + 6$, or $x = 10$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of y, not x.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.37 — B

Choice B is correct. A solution to a system of equations in the xy-plane is a point x, y that lies on the graph of each equation in the system. The first equation given is $x = 5$. Substituting 5 for x in the second given equation yields $y = 5 - 8$, or $y = -3$. It follows that in the xy-plane, the point $5, -3$ lies on the graph of each equation in the system. Therefore, the solution to the given system of equations in the xy-plane is $5, -3$.

Choice A is incorrect. The point $0, 0$ doesn't lie on the graph of either equation in the given system.

Choice C is incorrect. The point $5, -8$ doesn't lie on the graph of the second equation in the given system.

Choice D is incorrect. The point $5, 8$ doesn't lie on the graph of the second equation in the given system.

▼ 1.38 — B

Choice B is correct. Adding the second equation in the given system to the first equation in the given system yields $3x + (-3x + y) = 12 + -6$, which is equivalent to $y = 6$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.39 — D

Choice D is correct. Let l represent the cost, in dollars, of 1 loaf of bread, and let d represent the cost, in dollars, of 1 dozen eggs. It's given that the first customer paid 12.45 dollars for 1 loaf of bread and 2 dozen eggs. Therefore, the first customer's purchase can be represented by the equation $l + 2d = 12.45$. It's also given that the second customer paid 19.42 dollars for 4 loaves of bread and 1 dozen eggs. Therefore, the second customer's purchase can be represented by the equation $4l + d = 19.42$. The equations $l + 2d = 12.45$ and $4l + d = 19.42$ form a system of linear equations, which can be solved by elimination to find the value of d. Multiplying the first equation in the system by -4 yields $-4l - 8d = -49.8$. Adding $-4l - 8d = -49.8$ to the second equation, $4l + d = 19.42$, yields $-4l + 4l + -8d + d = -49.8 + 19.42$, which is equivalent to $-7d = -30.38$. Dividing both sides of this equation by -7 yields $d = 4.34$. Therefore, the cost, in dollars, of 1 dozen eggs is 4.34.

Choice A is incorrect. This is the cost, in dollars, of 1 loaf of bread.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

▼ 1.40 — B

Choice B is correct. Each given equation is written in slope-intercept form, $y = mx + b$, where m is the slope and $0, b$ is the y-intercept of the graph of the equation in the xy-plane. The graphs of two lines that have different slopes will intersect at exactly one point. The graph of the first equation is a line with slope 1. The graph of the second equation is a line with slope 8. Since the graphs are lines with different slopes, they will intersect at exactly one point.

Choice A is incorrect because two graphs of linear equations have 0 intersection points only if they are parallel and therefore have the same slope.

Choice C is incorrect because two graphs of linear equations in the xy-plane can have only 0, 1, or infinitely many points of intersection.

Choice D is incorrect because two graphs of linear equations in the xy-plane can have only 0, 1, or infinitely many points of intersection.

▼ 1.41 — 1

The correct answer is 1. Subtracting the second equation from the first equation in the given system of equations yields $3x - 3x + 6 - 4 = 4y - 2y$, which is equivalent to $0 + 2 = 2y$, or $2 = 2y$

. Dividing each side of this equation by 2 yields $1 = y$.

▼ 1.42 — D

Choice D is correct. Multiplying both sides of $y = 2x - 3$ by 5 results in $5y = 10x - 15$. Multiplying both sides of $3y = 5x$ by 2 results in $6y = 10x$. Subtracting the resulting equations yields $5y - 6y = (10x - 15) - (10x)$, which simplifies to $-y = -15$. Dividing both sides of $-y = -15$ by -1 results in $y = 15$.

Choices A and B are incorrect and may result from incorrectly subtracting the transformed equation. Choice C is incorrect and may result from finding the value of x instead of the value of y.

▼ 1.43 — A

Choice A is correct. A system of two linear equations in two variables, x and y, has zero points of intersection if the lines represented by the equations in the xy-plane are distinct and parallel. The graphs of two lines in the xy-plane represented by equations in slope-intercept form, $y = mx + b$, are distinct if the y-coordinates of their y-intercepts, b, are different and are parallel if their slopes, m, are the same. For the two equations in the given system, $y = 2x + 10$ and $y = 2x - 1$, the values of b are 10 and -1, respectively, and the values of m are both 2. Since the values of b are different, the graphs of these lines have different y-coordinates of the y-intercept and are distinct. Since the values of m are the same, the graphs of these lines have the same slope and are parallel. Therefore, the graphs of the given equations are lines that intersect at zero points in the xy-plane.

Choice B is incorrect. The graphs of a system of two linear equations have exactly one point of intersection if the lines represented by the equations have different slopes. Since the given equations represent lines with the same slope, there is not exactly one intersection point.

Choice C is incorrect. The graphs of a system of two linear equations can never have exactly two intersection points.

Choice D is incorrect. The graphs of a system of two linear equations have infinitely many intersection points when the lines represented by the equations have the same slope and

the same y-coordinate of the y-intercept. Since the given equations represent lines with different y-coordinates of their y-intercepts, there are not infinitely many intersection points.

▼ 1.44 — 25

The correct answer is 25. Subtracting the second equation from the first equation in the given system of equations yields $2a - 2a + 8b - 4b = 198 - 98$, which is equivalent to $0 + 4b = 100$, or $4b = 100$. Dividing each side of this equation by 4 yields $b = 25$.

▼ 1.45 — C

Choice C is correct. The graph of a system of equations is the graph that shows the lines represented by each of the equations in the system. The x-intercept of the graph of each given equation can be found by substituting 0 for y in each equation: $x+5(0)=5$, or $x=5$, and $2x-0=-4$, or $x=-2$. The y-intercept of the graph of each equation can be found by substituting 0 for x in each equation: $0+5y=5$, or $y=1$, and $2(0)-y=-4$ or $y=4$. Using these x- and y- intercept values, the line that has equation $x+5y=5$ passes through the points $(0,1)$ and $(5,0)$, and the line that has equation $2x-y=-4$ passes through the points $(0,4)$ and $(-2,0)$. Only the lines in choice C pass through these points and can be used to solve the given system of equations.

Choices A, B, and D are incorrect. In choices A and B, neither line passes through $(0,1)$ and $(5,0)$ or $(0,4)$ and $(-2,0)$. In choice D, although one line passes through $(0,4)$ and $(-2,0)$ the other line doesn't pass through $(0,1)$ and $(5,0)$.

▼ 1.46 — A

Choice A is correct. It's given that the sum of a number x and 7 is twice as large as a number y. This can be described by the equation $x + 7 = 2y$. It's also given that the number y is 3 less than the number x. This can be described by the equation $y = x - 3$. Therefore, the system consisting of the equations $x + 7 = 2y$ and $y = x - 3$ describes this situation.

Choice B is incorrect. The equation $y = 3 - x$ describes a situation where the number y is x less than 3.

Choice C is incorrect. The equation $2x + 7 = y$ describes a situation where the number y is twice the sum of a number x and 7.

Choice D is incorrect. The equation $2x + 7 = y$ describes a situation where the number y is twice the sum of a number x and 7, and the equation $y = 3 - x$ describes a situation where a number y is x less than 3.

▼ 1.47 — C

Choice C is correct. It's given by the first equation in the system that $y = 3x$. Substituting $3x$ for y in the equation $2x + y = 12$ yields $2x + 3x = 12$, or $5x = 12$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.48 — C

Choice C is correct. Adding the second equation in the given system to the first equation in the given system yields $x + 3 + x - 3 = -2y + 5 + 2y + 7$. Adding like terms in this equation yields $2x = 12$. Thus, the value of $2x$ is 12.

Choice A is incorrect. This is the value of y, not $2x$.

Choice B is incorrect. This is the value of x, not $2x$.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.49 — C

Choice C is correct. It's given that m represents the original price, in dollars, of the mirror, and v represents the original price, in dollars, of the vase. It's also given that the combined original price for the mirror and the vase is $\$60$. This can be represented by the equation $m + v = 60$. After a 25% discount to the mirror is applied, the sale price of the mirror is 75% of its original price. This can be represented by the expression $0.75m$. After a 45% discount to the vase is applied, the sale price of the vase is 55% of its original price. This can be represented by the expression $0.55v$. It's given that the combined sale price for the two items is $\$39$. This can be represented by the equation $0.75m + 0.55v = 39$. Therefore, the

system of equations consisting of the equations $m + v = 60$ and $0.75m + 0.55v = 39$ gives the original price m, in dollars, of the mirror and the original price v, in dollars, of the vase.

Choice A is incorrect. The second equation in this system of equations represents a 45% discount to the mirror and a 25% discount to the vase.

Choice B is incorrect. The second equation in this system of equations represents a 55% discount to the mirror and a 75% discount to the vase.

Choice D is incorrect. The second equation in this system of equations represents a 75% discount to the mirror and a 55% discount to the vase.

▼ 1.50 — D

Choice D is correct. It's given that a wire with a length of 106 inches is cut into two parts. It's also given that one part has a length of x inches and the other part has a length of y inches. This can be represented by the equation $x + y = 106$. It's also given that the value of x is 6 more than 4 times the value of y. This can be represented by the equation $x = 4y + 6$. Substituting $4y + 6$ for x in the equation $x + y = 106$ yields $4y + 6 + y = 106$, or $5y + 6 = 106$. Subtracting 6 from each side of this equation yields $5y = 100$. Dividing each side of this equation by 5 yields $y = 20$. Substituting 20 for y in the equation $x = 4y + 6$ yields $x = 4(20) + 6$, or $x = 86$.

Choice A is incorrect. This value represents less than half of the total length of 106 inches; however, x represents the length of the longer part of the wire, since it's given that the value of x is 6 more than 4 times the value of y.

Choice B is incorrect. This value represents less than half of the total length of 106 inches; however, x represents the length of the longer part of the wire, since it's given that the value of x is 6 more than 4 times the value of y.

Choice C is incorrect. This represents a part that is 6 more than the length of the other part, rather than 6 more than 4 times the length of the other part.

▼ 1.51 — C

Choice C is correct. Since the graph of $y = x + 3$ intersects the graph of $y = 2x - 6$ at the point (a,b), the ordered pair (a,b) is the solution to the system of linear equations consisting of $y = x + 3$ and $y = 2x - 6$, and the value of a is the value of x in the solution of this system.

Since both $x+3$and $2x-6$ are equal to y, it follows that $x+3=2x-6$. Subtracting x from and adding 6 to both sides of the equation yields $9=x$. Therefore, the value of a is 9.

Choices A and B are incorrect and may result from a calculation or conceptual error in solving the system of equations consisting of $y=x+3$ and $y=2x-6$. Choice D is incorrect. This is the value of b, not a.

▼ 1.52 — B

Choice B is correct. The intercepts of the graph of each equation can be found by substituting 0 for one variable at a time. For the equation $x + y = 53$, substituting 0 for y yields $x = 53$, so the x-intercept of the graph of the equation is $(53, 0)$. Substituting 0 for x in this equation yields $y = 53$, so the y-intercept of the graph of the equation is $(0, 53)$. Thus, the graph of the equation $x + y = 53$ has intercepts at $(53, 0)$ and $(0, 53)$. For the equation $11x + 18y = 730$, substituting 0 for y yields $x = \frac{730}{11}$, so the x-intercept of the graph of the equation is approximately $(66, 0)$. Substituting 0 for x in this equation yields $y = \frac{730}{18}$, so the y-intercept of the graph of the equation is approximately $(0, 41)$. Thus, the graph of the equation $11x + 18y = 730$ has intercepts at approximately $(66, 0)$ and $(0, 41)$. Choice B is a graph that shows one line with intercepts at $(0, 53)$ and $(53, 0)$ and another line with intercepts at approximately $(0, 41)$ and $(66, 0)$. Thus, choice B represents this situation.

Choice A is incorrect. This graph represents the equations $y = x$ and $11x + 18y = 730$.

Choice C is incorrect. This graph represents the equations $x + y = 53$ and $-11x + 18y = 730$.

Choice D is incorrect. This graph represents the equations $y = 53$ and $-11x + 18y = 730$.

▼ 1.53 — 40

The correct answer is 40. It's given in the first equation of the system that $y = -2x$. Substituting $-2x$ for y in the second equation of the system yields $3x + -2x = 40$. Combining like terms on the left-hand side of this equation yields $x = 40$. Therefore, the value of x is 40.

▼ 1.54 — B

Choice B is correct. If the bus traveled at an average speed of 55 miles per hour mph on the highway for x hours, then the bus traveled $55x$ miles on the highway. If the bus traveled at an average speed of 25 mph on local roads for y hours, then the bus traveled $25y$ miles on local roads. It's given that the trip was 160 miles. This can be represented by the equation $55x + 25y = 160$. It's also given that the trip took 4 hours. This can be represented

by the equation $x + y = 4$. Therefore, the system consisting of the equations $55x + 25y = 160$ and $x + y = 4$ represents this situation.

Choice A is incorrect. This system of equations represents a situation where the trip was 4 miles and took 160 hours.

Choice C is incorrect. This system of equations represents a situation where the trip was 4 miles and took 160 hours, and the bus traveled at an average speed of 25 mph on the highway and 55 mph on local roads.

Choice D is incorrect. This system of equations represents a situation where the bus traveled at an average speed of 25 mph on the highway and 55 mph on local roads.

▼ 1.55 — 3

The correct answer is 3. It's given that $y = 9x + 12$. Substituting $9x + 12$ for y in the second equation in the system, $x + 7y = 20$, yields $x + 79x + 12 = 20$, which gives $x + 63x + 84 = 20$, or $64x + 84 = 20$. Subtracting 84 from each side of this equation yields $64x = -64$. Dividing each side of this equation by 64 yields $x = -1$. Substituting -1 for x in the first equation in the system, $y = 9x + 12$, yields $y = 9\text{-}1 + 12$, or $y = 3$. Therefore, the value of y is 3.

▼ 1.56 — C

Choice C is correct. A line in the xy-plane that passes through the points (x_1, y_1) and (x_2, y_2) has slope m, where $m = \frac{y_2 - y_1}{x_2 - x_1}$. The equation of the line can be written in the form $y = mx + b$, where b is the y-intercept represented by $(0, b)$. One of the lines shown in the graph passes through the points $(0, 7)$ and $(11, 11)$. Substituting 0 for x_1, 7 for y_1, 11 for x_2, and 11 for y_2 in the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{11 - 7}{11 - 0}$, or $m = \frac{4}{11}$. Substituting $\frac{4}{11}$ for m and 7 for b in the equation $y = mx + b$ yields $y = \frac{4}{11}x + 7$. Therefore, an equation of this line is $y = \frac{4}{11}x + 7$. Similarly, the other line shown in the graph passes through the points $(0, 13)$ and $(11, 11)$. Substituting 0 for x_1, 13 for y_1, 11 for x_2, and 11 for y_2 in the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{11 - 13}{11 - 0}$, or $m = \frac{-2}{11}$. Substituting $\frac{-2}{11}$ for m and 13 for b in the equation $y = mx + b$ yields $y = \frac{-2}{11}x + 13$. Therefore, an equation of this line is $y = \frac{-2}{11}x + 13$. So, the functions f and g could be represented by $f(x) = \frac{-2}{11}x + 13$ and $g(x) = \frac{4}{11}x + 7$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.57 — B

Choice B is correct. Adding the first equation to the second equation in the given system yields $x + 2y + x - 2y = 6 + 4$, or $x + x + 2y - 2y = 10$. Combining like terms in this equation yields $2x = 10$. Dividing both sides of this equation by 2 yields $x = 5$. Thus, the value of x is 5.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $2x$, not x.

▼ 1.58 — C

Choice C is correct. It's given that Ellen bought two types of snack bags: chips and crackers. It's also given that each box of chips contains 24 bags. Thus, if Ellen bought x boxes of chips, the number of bags of chips is $24x$. Additionally, it's given that each box of chips costs $\$11$. Thus, if Ellen bought x boxes of chips, the cost for all the bags of chips is $11x$. It's also given that each box of crackers contains 16 bags. Thus, if Ellen bought y boxes of crackers, the number of bags of crackers is $16y$. Additionally, it's given that each box of crackers costs $\$9$. Thus, if Ellen bought y boxes of crackers, the cost for all the bags of crackers is $9y$. It's given that it cost Ellen $\$497$ to buy 968 bags. Since Ellen bought $24x$ bags of chips and $16y$ bags of crackers, the total number of snack bags that Ellen bought can be represented by the equation $24x + 16y = 968$. Additionally, since the cost for all the bags of chips is $11x$ and the cost for all the bags of crackers is $9y$, the total cost for all the snack bags that Ellen bought can be represented by the equation $11x + 9y = 497$. Therefore, the system consisting of the equations $11x + 9y = 497$ and $24x + 16y = 968$ represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This system of equations represents a situation where the total cost is $\$968$, not $\$497$, and the total number of bags is 497, not 968.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.59 — D

Choice D is correct. Multiplying the second equation in the given system by 2 yields $\frac{10}{2}x + 12y = 46$. Adding this equation to the first equation in the system yields

$\frac{7}{2}x + 6y + \frac{10}{2}x + 12y = 25 + 46$, which is equivalent to $\frac{7}{2}x + \frac{10}{2}x + 6y + 12y = 25 + 46$, or $\frac{17}{2}x + 18y = 71$. Therefore, the value of $\frac{17}{2}x + 18y$ is 71.

Choice A is incorrect. This is the value of x, not the value of $\frac{17}{2}x + 18y$.

Choice B is incorrect. This is the value of y, not the value of $\frac{17}{2}x + 18y$.

Choice C is incorrect. This the value of $\frac{7}{2}x + 6y + \frac{5}{2}x + 6y$, or $6x + 12y$, not the value of $\frac{17}{2}x + 18y$.

▼ 1.60 — A

Choice A is correct. It's given that a radio show stated that 3 times as many people voted in favor of the proposal as people who voted against it. Let x represent the number of people who voted against the proposal. It follows that $3x$ is the number of people who voted in favor of the proposal and $3x - x$, or $2x$, is how many more people voted in favor of the proposal than voted against it. It's also given that a social media post reported that $15{,}000$ more people voted in favor of the proposal than voted against it. Thus, $2x = 15{,}000$. Since $2x = 15{,}000$, the value of x must be half of $15{,}000$, or $7{,}500$. Therefore, $7{,}500$ people voted against the proposal.

Choice B is incorrect. This is how many more people voted in favor of the proposal than voted against it, not the number of people who voted against the proposal.

Choice C is incorrect. This is the number of people who voted in favor of the proposal, not the number of people who voted against the proposal.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.61 — A

Choice A is correct. The given system of linear equations can be solved by the elimination method. Multiplying each side of the second equation in the given system by 3 yields $2x + 2y(3) = 103$, or $6x + 6y = 30$. Subtracting this equation from the first equation in the given system yields $6x + 7y - 6x + 6y = 28 - 30$, which is equivalent to $6x - 6x + 7y - 6y = 28 - 30$, or $y = -2$.

Choice B is incorrect. This is the value of x, not the value of y.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

▼ 1.62 — 24

The correct answer is 24. The given system of equations can be solved by the substitution method. The first equation in the given system of equations is $y = \frac{1}{3}x - 14$. Substituting $\frac{1}{3}x - 14$ for y in the second equation in the given system yields $\frac{1}{3}x - 14 = -x + 18$. Adding 14 to both sides of this equation yields $\frac{1}{3}x = -x + 32$. Adding x to both sides of this equation yields $\frac{4}{3}x = 32$. Multiplying both sides of this equation by $\frac{3}{4}$ yields $x = 24$.

▼ 1.63 — D

Choice D is correct. It's given that the system has infinitely many solutions. A system of two linear equations has infinitely many solutions when the two linear equations are equivalent. When one equation is a multiple of another equation, the two equations are equivalent. Multiplying each side of the given equation by 2 yields $2y = 26x + 3$. Thus, $2y = 26x + 3$ is equivalent to the given equation and could be the second equation in the system.

Choice A is incorrect. The system consisting of this equation and the given equation has one solution rather than infinitely many solutions.

Choice B is incorrect. The system consisting of this equation and the given equation has one solution rather than infinitely many solutions.

Choice C is incorrect. The system consisting of this equation and the given equation has no solutions rather than infinitely many solutions.